# Resolution Geometry: GR Automation

## Observable Sectors, Transport and Source on a Gravitational Background: a Locked Pipeline with Structural Proofs

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**Abstract**

A finite observer in a gravitational field sees a field through a detector. Which part of the field is resolved, how that part moves when the geometry changes, and which source the observer may present to a gravitational equation are usually decided by hand. This print makes each decision a theorem. The pipeline is

$$(M, g) + (B, \Sigma, S) \longrightarrow (F, H_g) \longrightarrow (P_R, P_N) \longrightarrow A_{RN} \longrightarrow W_{\mathrm{obs}} \longrightarrow T^{\mathrm{obs}}_{\mu\nu},$$

with detector $B$, noise covariance $\Sigma$, action $S$, Fisher form $F = B^{\top}\Sigma^{-1}B$ and energy form $H_g$.

*Structure.* The null sector is $\ker B$. The energy form of the metric supplies its complement, which Resolution Geometry had left as a declaration. The resolved sector is the span of the Green functions of the detector, and the health spectrum is the spectrum of $\Sigma^{-1}G$, with $G$ the detector Green matrix. That spectrum is a complete invariant of the pair of forms on the observable quotient $V/\ker B$. It is independent of field coordinates and of the grid, provided the detector functionals are bounded on the energy space; point probes in dimension two or more are not. The groupoid of the hole and the quotient agree up to Morita equivalence.

*Transport.* The derivative of the resolved projector is purely the null-from-resolved block, $\mathrm{d}P_R = -P_N H^{-1}(\mathrm{d}H)P_R$, and this block is the mixed matrix element of the stress tensor. The full sector has exactly trivial holonomy over the space of metrics; holonomy arises only from a truncated budget or a moving detector. Isometries leave the health spectrum invariant, and the rate of change along any flow is the stress contracted with the deformation tensor. That is where the Levi-Civita connection enters.

*Source.* $W_{\mathrm{obs}}$ is the quotient norm of the energy: the least energy compatible with the readings. Its metric variation is the stress tensor of the eliminated field, exactly, when the readings are held fixed; the identity fails in metric-dependent frames, an omitted step we correct. The true energy exceeds $W_{\mathrm{obs}}$ by the energy of the hole, so the deficit has one sign. $T^{\mathrm{obs}}$ is conserved off the detector and exchanges momentum with it; Bianchi consistency therefore requires the detector in the ledger.

*Exact example.* For shell detectors around a Schwarzschild mass, $G_{ij} = \min(u_i, u_j)/(8\pi M)$ with $u = -\ln(1 - 2M/r)$. This gives a capacitance law, a horizon wall, and a closed-form transport amplitude that replaces an earlier hand-seeded value.

A battery of 17 groups, each with a canary, checks every arrow on computed objects; twelve mutations are caught. The print does not derive general relativity, does not choose the detector, and does not recover what lies in the hole.

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# 1 Purpose, setting and status

## 1.1 What is automated

Resolution Geometry (RG) studies what a declared family of observation protocols can distinguish. In a gravitational setting three objects were assigned by hand in earlier work: the split of a field into a resolved and a null part, the transport of that split when the geometry changes, and the source tensor the observer attributes to what it sees. *Automation* means that each is generated from the declared data $(M, g; B, \Sigma, S)$ by a construction that is forced, natural, and independent of discretization.

A peer can object that such a pipeline is only finite matching: numbers that agree because they were built to agree. Section 10 lists these objections and the theorem or canary that answers each. Two tables in the earlier automation notes were of this kind, and Section 12 withdraws them.

## 1.2 Setting

**Definition 1.1** (Static setting)**.** Let $(M, g)$ contain a static region: $\mathrm{d}s^2 = -N^2\mathrm{d}t^2 + h_{ij}\mathrm{d}x^i\mathrm{d}x^j$ on $\mathbb{R} \times \Sigma_0$ with lapse $N > 0$ and Riemannian $h$, both independent of $t$. A scalar field with

action $S = -\frac{1}{2}\int \sqrt{-g}\,(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + m^2\phi^2)$ has, on static configurations, $S = -\int \mathrm{d}t\, E_g[\phi]$ with the Killing energy

$$E_g[\phi] = \tfrac{1}{2}H_g(\phi,\phi), \qquad H_g(u,v) = \int_{\Sigma_0} N\sqrt{h}\,(h^{ij}\partial_i u\,\partial_j v + m^2 uv)\mathrm{d}^3x. \tag{1}$$

The field space $V$ is the completion of test functions under $H_g$ (with a boundary condition or $m > 0$ making $H_g$ positive definite). A *detector* is a linear map $B : V \to \mathbb{R}^k$ whose components $b_i$ are bounded on $V$, with noise covariance $\Sigma \succ 0$. The Fisher form is $F = B^\top\Sigma^{-1}B$.

Time-dependent fields on the same background are treated in phase space (Proposition 5.6). Everything in Sections 3–6 uses only a positive definite form $H$ depending on a parameter $g$, and a bounded $B$. The static scalar is the worked model; the theorems are stated for the general pair.

### 1.3 Status vocabulary and non-claims

**proved** Proved here; the battery group is given in brackets.

**identified** Coincides with an established theorem, which is cited. No priority is claimed.

**imported** Used from the literature without reproof.

**declared** An input the construction cannot supply.

- *No derivation of general relativity.* The background $(M, g)$ is given. The companion *Tensor Derivation* shows that the metric response is the only admissible source object and selects the Einstein branch; this print uses that result and does not reprove it.
- *No choice of detector.* $B$ and $\Sigma$ are DECLARED.
- *No recovery from the hole.* Directions in $\ker B$ are reported, never reconstructed.

## 2 Layer 0: the positive slot

RG needs a positive definite form to split a space orthogonally. A Lorentzian metric cannot fill that slot, and neither can the Hessian of a Lorentzian action.

**Proposition 2.1** (Lorentzian wall)**.** [GM] *Let $H$ be an indefinite nondegenerate symmetric form and $F \succeq 0$.*

1. *The relative eigenvalues of $(F, H)$ are real but take both signs.*
2. *There is a detector $b$ whose representer $H^{-1}b$ is $H$-null. For it $b^\top H^{-1}b = 0$, the operator $H^{-1}bb^\top$ is nilpotent and nonzero, and no $H$-orthogonal projector onto the resolved line exists.*

*Proof.* The nonzero eigenvalues of $H^{-1}F$ are those of the symmetric matrix $F^{1/2}H^{-1}F^{1/2}$, which is indefinite. For (2) take $v$ with $H(v,v) = 0$ and $b = Hv$. Then $(H^{-1}bb^\top)^2 = H^{-1}b\,(b^\top H^{-1}b)\,b^\top = 0$. The battery finds 15 negative and 21 positive directions on a $6 \times 6$ spacetime grid, a null vector, the nilpotent block, and up to 4 negative relative eigenvalues for random detectors. □

General relativity therefore enters the positive slot through the Killing energy (1) of a static region, or through the Hamiltonian on phase space. This is not a convenience. It is the statement that a resolved/null split needs a conserved positive form, and that only a stationary observer has one canonically.

*Remark* 2.2 (Walls where the form must be declared). At a Killing horizon $N \to 0$ and $H_g$ degenerates (Section 7). In an ergoregion the Killing energy is indefinite. On a non-stationary background no conserved positive form exists; a positive form then amounts to the choice of a slicing or of a quasi-free state [9, 10], and is DECLARED. This parallels the no-go for a time-independent radar metric on non-stationary slices in the main print (L106).

# 3 Layer I: the observation pencil

## 3.1 Invariants and canonical sectors

**Theorem 3.1** (The health spectrum is a complete invariant). [GA] *Let $H \succ 0$ and $F \succeq 0$ on a finite-dimensional $V$. The eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$ of $Fv = \lambda Hv$ are invariant under every linear change of field coordinates, $(F, H) \mapsto (T^\top FT, T^\top HT)$. Two pairs are related by such a change iff their spectra agree.*

*Proof.* $T^\top FTw = \lambda T^\top HTw$ iff $F(Tw) = \lambda H(Tw)$. Conversely, simultaneous diagonalization gives bases with $V_a^\top H_a V_a = I$ and $V_a^\top F_a V_a = \Lambda$; then $T = V_1 V_2^{-1}$ carries one pair to the other. IDENTIFIED: simultaneous diagonalization of a definite pencil. □

The eigenvalue $\lambda_i$ is the Fisher information gained per unit of energy spent in mode $i$. It has a meaning (Proposition 3.7) and does not depend on how the field is parametrized.

**Theorem 3.2** (Canonical sectors). [GA] *With full budget $k = \operatorname{rank} B$:*

1. *the null sector is $N = \ker B$, independent of $g$ and of $\Sigma$;*
2. *the resolved sector is $R = N^{\perp_H}$, with $H$-orthogonal projector*

$$P_R = H^{-1}B^\top (BH^{-1}B^\top)^{-1}B, \qquad P_N = 1 - P_R; \tag{2}$$

3. *$P_R$ is independent of $\Sigma$. The noise covariance only orders the resolved modes.*

*For a budget $k' < k$, the span $R_{k'}$ of the top $k'$ eigenvectors maximizes $\operatorname{tr}(Q^\top FQ)$ over $H$-orthonormal $k'$-frames (IDENTIFIED: Ky Fan [1]). $R_{k'}$ depends on $g$ and on $\Sigma$.*

*Proof.* Eigenvectors with $\lambda > 0$ span $H^{-1}\operatorname{range} F = H^{-1}\operatorname{range} B^\top$, which is $(\ker B)^{\perp_H}$ for every $\Sigma \succ 0$. Formula (2) is the $H$-orthogonal projector with kernel $\ker B$. □

*Remark* 3.3 (The metric supplies the complement). The main print shows that the null kernel is determined by the law while its complement is not: the coupling is complement-relative (L117). Theorem 3.2 is the physical resolution. The hole is fixed by the detector alone, and the energy form of the metric selects the complement. This is the precise content of "placing general relativity in RG".

*Remark* 3.4 (Capture identities). The captured fraction $\sum_{i \leq k'} \lambda_i / \sum_i \lambda_i$ equals 1 exactly when $k' \geq \operatorname{rank} B$, and a random $H$-orthonormal $k'$-frame captures $k'/n$ in expectation (battery: $0.2169 \pm 0.0016$ against $3/14 = 0.2143$). These are identities. A table reporting capture 1 against a random baseline $k'/n$ certifies nothing about a model.

## 3.2 Representers and the Green matrix

**Theorem 3.5** (Representer theorem). [GC] *Let $r_i = H^{-1}b_i$ be the Riesz representer of $b_i$, so that $H(r_i, v) = b_i(v)$. Then:*

1. *$R = \operatorname{span}\{r_1, \ldots, r_k\}$;*

2. *the nonzero health spectrum is the spectrum of the $k \times k$ matrix $\Sigma^{-1}G$, where $G_{ij} = b_i(H^{-1}b_j)$;*

3. *the least-energy field with readings $y$ is $\phi^\star = \sum_i \mu_i r_i$ with $\mu = G^{-1}y$, and it solves the field equation with sources on the detector: $H\phi^\star = B^\top \mu$.*

*Proof.* (1) is Theorem 3.2. For (2), $H^{-1}B^\top\Sigma^{-1}B$ and $\Sigma^{-1}BH^{-1}B^\top$ have the same nonzero eigenvalues. (3) is the Lagrange condition for $\min\{\frac{1}{2}H(\phi,\phi) : B\phi = y\}$. IDENTIFIED: Riesz; the representer theorem of spline and kernel theory [2]. □

In the static setting $H_g^{-1}$ is the Green operator $\mathcal{G}_g$ of the elliptic operator $-D^i(ND_i\,\cdot) + Nm^2$. The resolved sector is the set of fields that sources placed on the detector would generate, and $G_{ij}(g) = \langle b_i, \mathcal{G}_g b_j\rangle$ is the mutual response of detectors $i$ and $j$. The geometry enters the observation layer only through its Green function.

**Theorem 3.6** (Continuum limit and admissibility)**.** [GD] *If every $b_i$ is bounded on the energy space, then $G$ and the health spectrum are defined in the continuum, and consistent discretizations converge to them. If some $b_i$ is unbounded, $G_{ii} = \infty$ and the spectrum of any discretization is a mesh artefact. A point evaluation is bounded in one space dimension and unbounded in dimension two or more.*

*Proof.* $G_{ii} = \|b_i\|^2_{V^*}$. Point evaluation is bounded on $H^1$ iff the dimension is one (Sobolev). The battery gives: in one dimension, convergence to $G(x,y) = \sinh(mx_<)\sinh(m(L - x_>))/(m\sinh mL)$ at orders 1.9997 and 1.9999; in two dimensions, block averages converge (0.2489, 0.2431, 0.2417), while the point self-energy grows by 0.1107 and 0.1104 per mesh halving, against $\ln 2/2\pi = 0.1103$. □

This is the answer to the objection of finite matching. The pipeline is a statement about Green functions. A grid is admissible exactly when it approximates them, and the wall where it cannot is located.

### 3.3 Meaning and monotonicity

**Proposition 3.7** (Identification with Bayesian inference)**.** [GB] *Give the field the Gibbs prior of the energy form, with covariance $(\beta H)^{-1}$. Then the posterior covariance is $(F + \beta H)^{-1}$, the mutual information between field and data is $\frac{1}{2}\sum_i \ln(1 + \lambda_i/\beta)$, and the top generalized eigenvectors give the optimal low-rank update of the prior covariance.*

*Proof.* Whitening by $H^{1/2}$ reduces to a standard Gaussian prior. IDENTIFIED: optimal low-rank posterior approximation and the likelihood-informed subspace [3, 4]. The battery matches the mutual information to rounding. The optimal rank-3 update has Förstner distance 0.875 to the exact posterior; the best of 200 random frames has 3.23, and the wrong ordering 4.05. □

The prior precision of that literature is here the energy form. It is not chosen by the analyst; the metric supplies it.

**Proposition 3.8** (Monotone functor of the experiment)**.** [GQ] *Adding an independent detector row raises every $\lambda_i$ and enlarges $R$. Coarse-graining the data, $y \mapsto Ay$, lowers $F$ in the positive semidefinite order and lowers every $\lambda_i$. $W_{\text{obs}}$ rises under added constraints.*

*Proof.* Weyl's inequalities for a positive semidefinite update; the data-processing inequality for Fisher information (main print, L35); a minimum over a smaller set. □

# 4 Structure: the observable quotient

**Theorem 4.1** (Two forms on the quotient)**.** [GJ] *Let $Q = V/\ker B$.*

1. *$F$ descends to a positive definite form $\bar{F}$ on $Q$.*
2. *The quotient norm of $H$ defines a positive definite form $\bar{H}$ on $Q$, with $\frac{1}{2}\bar{H}([\phi],[\phi]) = \min\{\frac{1}{2}H(\psi,\psi) : \psi \in \phi + \ker B\}$.*
3. *The restriction of the quotient map to $R$ is an isometry $(R, H|_R) \to (Q, \bar{H})$.*
4. *The health spectrum is the relative spectrum of $(\bar{F}, \bar{H})$, and by Theorem 3.1 it is the complete invariant of the observable structure.*

*Proof.* $\ker B = \operatorname{rad} F$ gives (1). The minimizer over a coset is its $H$-orthogonal component, which lies in $R$; this gives (2) and (3). (4) follows. □

**Definition 4.2** (Observation action)**.** $W_{\mathrm{obs}}(y; g) = \min\{S_g[\phi] : B\phi = y\}$. For the static quadratic model, $W_{\mathrm{obs}} = \frac{1}{2}\bar{H}_g = \frac{1}{2}y^\top G(g)^{-1}y$.

So $W_{\mathrm{obs}}$ is not an auxiliary construction. It is the energy form of the metric, pushed down to the observable quotient. The quotient map is a metric submersion; $R$ is its horizontal space and $\ker B$ its vertical space.

**Theorem 4.3** (The groupoid of the hole meets the quotient)**.** [GP] *The hole $\ker B$ acts on $V$ by translation, freely and properly. Then:*

1. *the algebra of continuous functions of the data vanishing at infinity is $C_0(Q)$, whose Gelfand spectrum is $Q$;*
2. *the $C^*$-algebra of the action groupoid $V \rtimes \ker B$ is Morita equivalent to $C_0(Q)$ (*IMPORTED*: [5, 6]).*

The bottom-up object (the groupoid of indistinguishable fields) and the top-down object (the quotient) therefore carry the same representation theory, and $W_{\mathrm{obs}}$ lives on their meeting point. The finite witness in the battery takes $V = \mathbb{Z}_3^2$ and $B = (1\ 1)$. The groupoid algebra has dimension $27 = 3 \cdot 3^2$, a sum of three full matrix blocks, with centre of dimension 3, the number of data values. The canary is a non-free action with one fixed point: its centre has dimension 3 against 2 orbits. Where the action is not free the quotient alone is not enough, and stabilizer representations appear. This is the algebraic form of RG's walls. Where both paths apply, the result agrees with the Gelfand meeting point of the companion *RG Topology*.

# 5 Layer II: transport

## 5.1 The transport block

**Theorem 5.1** (The block is the whole derivative)**.** [GF] *Let $H$ depend differentiably on a parameter and let $B$ be fixed. For the full sector,*

$$\mathrm{d}P_R = -P_N H^{-1}(\mathrm{d}H)P_R, \qquad P_R\,\mathrm{d}P_R = 0, \qquad \mathrm{d}P_R\,P_N = 0. \tag{3}$$

*Hence $A_{RN} := P_N(\mathrm{d}P_R)P_R$ equals $\mathrm{d}P_R$: the resolved sector moves only into the hole.*

*Proof.* $P_R$ is a projector with fixed kernel $\ker B$, so $\mathrm{d}P_R\,P_N = 0$. Differentiating $P_R^2 = P_R$ then gives $P_R\,\mathrm{d}P_R = 0$. Differentiating (2) with $G = BH^{-1}B^\top$ gives $\mathrm{d}P_R = -H^{-1}\mathrm{d}H\,P_R + P_R H^{-1}\mathrm{d}H\,P_R$. □

**Theorem 5.2** (Transport–stress duality)**.** [GG] *In the static setting the variation of the energy form is the polarized stress tensor:*

$$\delta H_g(u,v) = 2\int_{\Sigma_0} \sqrt{h}\,\Big[\rho(u,v)\,\delta N + \tfrac{1}{2}N\,T_{ij}(u,v)\,\delta h^{ij}\Big]\mathrm{d}^3x, \tag{4}$$

*with* $\rho(u,v) = \frac{1}{2}(h^{ij}\partial_i u\partial_j v + m^2 uv)$ *and* $T_{ij}(u,v) = \partial_{(i}u\,\partial_{j)}v - h_{ij}\rho(u,v)$*. Consequently, for* $r \in R$ *and* $n \in \ker B$*,*

$$H\big(A_{RN}[\delta g]\,r,\,n\big) = -\delta H_g(r,n). \tag{5}$$

*The transport block is the resolved–null matrix element of the stress tensor, contracted with the metric variation.*

*Proof.* Vary (1): $\partial(N\sqrt{h}\,h^{ij})/\partial h^{ij}$ produces the trace term. The second statement follows from (3), since $P_N$ is $H$-self-adjoint and $H(H^{-1}\delta H\,r,n) = \delta H(r,n)$. The battery compares a finite-difference $\delta H$ with the quadrature of (4) for independent lapse and radial-metric perturbations (relative error $7\times10^{-5}$) and checks the duality to $10^{-12}$. Without the pressure term the identity fails by 0.88. □

This joins Layers II and III. The object that rotates the sector is the same stress bilinear that Layer III will produce as a source. $A_{RN} = 0$ for a metric variation iff that variation does not couple resolved and null fields through the stress tensor.

**Corollary 5.3** (Conformal rescaling)**.** [GF] *For a massless field in three space dimensions, the uniform rescaling* $h \mapsto a^2h$ *gives* $H \mapsto aH$ *and* $A_{RN} = 0$*. A mass term breaks the proportionality and gives* $A_{RN} \neq 0$ *(battery:* $5.6\times10^{-9}$ *against* 0.37*).*

## 5.2 Holonomy

**Theorem 5.4** (Holonomy criterion)**.** [GH] *Fix* $B$ *and let* $g$ *run over a loop of metrics. Transport the resolved sector by projection,* $r \mapsto P_R(g')r$*.*

1. *For the full sector the holonomy is exactly the identity, on every loop and for every discretization of the loop.*
2. *Holonomy can arise only if (i) the budget is truncated, so that the discarded part depends on* $g$ *and* $\Sigma$*, or (ii) the detector moves along the loop, so that* $\ker B$ *varies.*

*Proof.* For any $g'$ and any $r$, $r - P_R(g')r = P_N(g')r \in \ker B$. Each transport step therefore preserves the class in $Q = V/\ker B$, and on returning to $R(g_0)$, which maps isomorphically to $Q$, the vector is recovered. For (2): in case (i) the null part $R^{\perp}_{k'}$ is not fixed; in case (ii) $\ker B$ is a bundle over the loop. □

The battery, on a loop in mass and field mass $(M, m^2) = (1 + 0.3\cos t,\ 0.05 + 0.04\sin t)$ with five shell detectors, gives:

- full sector: defect $2\times10^{-14}$ with 12 steps and $8\times10^{-14}$ with 100;
- truncated to two of five modes: defect 0.0111, 0.0112, 0.0112 at 100, 400, 1600 steps, a converged nonzero holonomy;
- a retraced path: defect falling to $2.5\times10^{-5}$ (method canary);
- a detector line rotating by $\pi$: holonomy $-1$ (Möbius kernel bundle).

Case (ii) is the remainder class $[K_N]$ of the main print (L120). Theorem 5.4 is a diagnostic: holonomy reported by an analysis at full rank and fixed detector is an error.

### 5.3 Where the Levi-Civita connection enters

**Theorem 5.5** (Isometry covariance and deformation rate)**.** [GL] *Let $\Phi_s$ be the flow of a vector field $\xi$ on $\Sigma_0$ that preserves the boundary conditions, and move the detector by $b_i^{(s)}(\phi) = b_i(\phi \circ \Phi_s)$. Then*

$$\frac{\mathrm{d}G_{ij}}{\mathrm{d}s}\Big|_0 = 2\int_{\Sigma_0} \sqrt{h}\,\Big[N\,T_{ab}(r_i, r_j)\,D^a\xi^b - \rho(r_i, r_j)\,\xi^a D_a N\Big]\mathrm{d}^3x, \tag{6}$$

*where $D$ is the Levi-Civita connection of $h$. If $\xi$ is a Killing field of $h$ with $\xi(N) = 0$, then $G$ and the health spectrum are invariant, and the sectors are intertwined by $\Phi_s$. For a general parameter, $\mathrm{d}\lambda_i = v_i^\top(\mathrm{d}F - \lambda_i\,\mathrm{d}H)v_i$ with $H(v_i, v_i) = 1$.*

*Proof.* Moving the detector by $\Phi_s$ at fixed metric equals fixing the detector and replacing $g$ by $\Phi_s^* g$. Hence $\mathrm{d}G_{ij}/\mathrm{d}s = -\delta H(r_i, r_j)$ with $\delta N = \mathcal{L}_\xi N$ and $\delta h^{ab} = \mathcal{L}_\xi h^{ab} = -2D^{(a}\xi^{b)}$; insert (4). The last formula is Hellmann–Feynman for a definite pencil. Check: for a massless field in flat space and the dilation $\xi = x$, (6) gives $-G_{ij}$, the scaling $G \propto 1/a$ of a detector of size $a$. □

The battery translates a detector on a flat torus (spectrum shift $6 \times 10^{-16}$) and on a torus with a non-uniform conformal factor (shift 0.068), and checks Hellmann–Feynman to $10^{-10}$. It also checks (6) in closed form for a massive field on an interval with $\xi = \sin(\pi x/L)$: both sides equal 0.024952, with relative difference below $10^{-6}$; the opposite sign convention fails. The deformation tensor $D^{(a}\xi^{b)}$ is the only way the connection appears. General relativity is placed in RG as follows: isometries of the static structure are automorphisms of the resolution structure, and the failure of a flow to be an isometry is measured by stress contracted with deformation.

**Proposition 5.6** (Killing time and the dynamical hole)**.** [GL] *On phase space $z = (\phi, \pi)$ with Hamiltonian form $\mathcal{E} = \mathrm{diag}(H_g, 1)$ and flow $U_t$, let the observation be $y_j = BU_{t_j}z$ and $F = \sum_j U_{t_j}^\top B^\top \Sigma^{-1} B U_{t_j}$. Then:*

1. *$U_t$ preserves $\mathcal{E}$, and shifting all observation times by $s$ leaves the spectrum of $(F, \mathcal{E})$ invariant;*
2. *the hole is the unobservable subspace $\bigcap_t \ker(BU_t)$, invariant under the flow, and so is its $\mathcal{E}$-complement.*

*Proof.* $U_s^\top \mathcal{E} U_s = \mathcal{E}$ and $F \mapsto U_s^\top F U_s$. (2) is Kalman's observability decomposition [7] (IDENTIFIED); the complement is invariant because the flow is $\mathcal{E}$-orthogonal. Battery: window shift changes the spectrum by $4 \times 10^{-15}$; a stiffness drifting in time changes it by 0.033 (canary); a symmetric five-site chain observed at its centre has a hole of dimension 4, the antisymmetric modes. □

*Remark* 5.7. If the field is in thermal equilibrium its covariance is $\beta^{-1}\mathcal{E}^{-1}$. The pencil $(F, \mathcal{E})$ is then the operator of balanced truncation [8], with the equilibrium covariance in the role of the controllability Gramian.

## 6 Layer III: the observable source

### 6.1 The envelope identity and an omitted step

**Theorem 6.1** (Envelope identity, operational form)**.** [GI] *Let $W_{\mathrm{obs}}(y; g) = \min\{S_g[\phi] : B\phi = y\}$ with $B$ independent of $g$, and let the minimizer $\phi^\star(y, g)$ be differentiable. Then for every metric parameter,*

$$\frac{\partial W_{\mathrm{obs}}}{\partial g} = \frac{\partial S_g}{\partial g}\Big|_{\phi=\phi^\star}. \tag{7}$$

*More generally, for a constraint $C(g)\phi = c$ with multiplier $\mu$ defined by $\partial_\phi S = C^\top \mu$,*

$$\frac{\partial W}{\partial g} = \frac{\partial S_g}{\partial g}\Big|_{\phi^\star} - \mu^\top \frac{\partial C}{\partial g}\,\phi^\star. \tag{8}$$

*Proof.* Differentiate the Lagrangian $S_g[\phi] - \mu^\top(C(g)\phi - c)$ at the stationary point. □

**Proposition 6.2** (Frame dependence; the omitted step)**.** [GI] *Hold fixed, instead of the readings, the $H_g$-orthonormal resolved coordinates $c = V_R(g)^\top H_g \phi$. Then $W = \frac{1}{2}|c|^2$ for every metric, $\partial W/\partial g = 0$, and the source would vanish identically.*

"Eliminate the null field at fixed resolved field" is therefore ambiguous. The earlier statement of the envelope identity did not say in which frame the resolved field is held. The readings $y$ are the only frame that is both operational and independent of the metric. The battery confirms the identity with fixed readings for the quadratic action (relative error $10^{-9}$) and for a quartic action (relative error $1.4 \times 10^{-9}$, Newton stationarity $8 \times 10^{-15}$). In the orthonormal frame the derivative is $-5 \times 10^{-7}$ against a naive envelope value of $-142$.

**Corollary 6.3** (Automated source)**.** *With readings fixed, $T^{\mathrm{obs}}_{\mu\nu} := -\frac{2}{\sqrt{-g}}\,\delta W_{\mathrm{obs}}/\delta g^{\mu\nu} = T_{\mu\nu}[\phi^\star]$: the stress tensor of the least-energy field compatible with the readings. In the static setting, $\delta W_{\mathrm{obs}} = \int \sqrt{h}\,[\rho^\star \delta N + \frac{1}{2} N T^\star_{ij} \delta h^{ij}]$.*

By the *Tensor Derivation*, the metric response is the unique source object compatible with relabeling invariance and universal coupling. The last arrow of the pipeline is thus forced, up to the improvement terms recorded there, which belong to the declared action $S$.

## 6.2 The deficit has one sign

**Theorem 6.4** (Energy deficit)**.** [GJ] *For every field $\phi$ with readings $y = B\phi$,*

$$E_g[\phi] = W_{\mathrm{obs}}(y; g) + \tfrac{1}{2} H_g(P_N\phi, P_N\phi) \;\geq\; W_{\mathrm{obs}}(y; g), \tag{9}$$

*with equality iff $\phi$ has no component in the hole. Refining the detector raises $W_{\mathrm{obs}}$ monotonically. The cross term vanishes in the integrated energy but not in the local stress density.*

*Proof.* $H$-orthogonality of $R$ and $\ker B$; Proposition 3.8. Battery: a field of energy 4201.4 has $W_{\mathrm{obs}} = 141.8$, 218.2, 1443.9, 2859.1 for 1, 3, 7, 38 shell detectors; the split holds to $10^{-15}$. The local cross stress reaches 37 while its integral is $5 \times 10^{-13}$. A Euclidean split of the same subspace has a cross energy of 1.3% (canary). □

A finite observer therefore certifies a lower bound on the source. The missing part is not noise and not a fitting error. It is the energy of the hole, it is non-negative, and nothing in the readings bounds it from above.

## 6.3 Conservation and the detector

**Theorem 6.5** (Ward identity for the observable source)**.** [GK] *Let $\sigma := -D^j(N D_j \phi^\star) + N m^2 \phi^\star$ be the source density of the eliminated field; by Theorem 3.5 it is supported on the detector. Then*

$$D^j(N\,T^\star_{ij}) + \rho^\star D_i N = -\sigma\,\partial_i \phi^\star, \qquad \textit{that is} \qquad \nabla^\mu T^{\mathrm{obs}}_{\mu i} = -\frac{\sigma}{N}\,\partial_i \phi^\star. \tag{10}$$

*$T^{\mathrm{obs}}$ is conserved wherever the detector is absent. Across the detector it exchanges momentum at the rate of the force on the detector source.*

*Proof.* Expand $D^j(N T_{ij})$ with $T_{ij} = \partial_i\phi\partial_j\phi - h_{ij}\rho$. The terms with second derivatives cancel against $N D_i \rho$, leaving $[D^j(N D_j\phi) - N m^2 \phi]\partial_i\phi - \rho D_i N$. □

In one flat dimension $T_{xx} = \frac{1}{2}(\phi'^2 - m^2\phi^2)$ is piecewise constant, with jump $-\mu_i \phi'_{\mathrm{avg}}(x_i)$ at detector $i$. The battery finds a variation between detectors of $4 \times 10^{-7}$ and $1 \times 10^{-7}$ on two grids (second order), a jump-to-force error of $6 \times 10^{-7}$, and a global balance to $8 \times 10^{-8}$. Adding a null-sector bump produces a variation of 3.3 (canary).

**Corollary 6.6** (Bianchi consistency)**.** *The equation $G_{\mu\nu} = 8\pi G\, T^{\mathrm{obs}}_{\mu\nu}$ is inconsistent with the contracted Bianchi identity on the support of the detector. Consistency requires the stress of the detector in the source: the observer must be in the ledger.*

This is not a defect of the construction. It records what the construction is: $\phi^\star$ is the field the detector itself would source, and a source without its support is not conserved.

# 7 Exact example: shell detectors around a Schwarzschild mass

Take $N^2 = 1 - 2M/r$, $h = N^{-2}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$, a massless field, and detectors that read the angular average of $\phi$ on spheres of areal radius $r_i > 2M$. Fields with no monopole part lie in the hole. On the monopole sector

$$E = \tfrac{1}{2}\int_{2M}^{\infty} 4\pi r(r-2M)\,\phi'^2\,\mathrm{d}r = \tfrac{1}{2}\cdot 8\pi M \int_0^{\infty} \Big(\frac{\mathrm{d}\phi}{\mathrm{d}u}\Big)^2 \mathrm{d}u, \qquad u = -\ln\Big(1 - \frac{2M}{r}\Big). \tag{11}$$

In the coordinate $u$ the exterior is a uniform string. Infinity is at $u = 0$ and the horizon at $u = \infty$.

**Theorem 7.1** (Green matrix, capacitance, horizon wall)**.** [GE]

1. $G_{ij} = \dfrac{\min(u_i, u_j)}{8\pi M} = -\dfrac{\ln(1 - 2M/\max(r_i, r_j))}{8\pi M}$*. This is the Wiener kernel in $u$; its inverse is tridiagonal.*

2. *A single reading $y$ at $r_0$ certifies the energy $W_{\mathrm{obs}} = \frac{1}{2}Cy^2$ with capacitance*

$$C(r_0, M) = \frac{8\pi M}{u_0} = \frac{8\pi M}{|\ln(1 - 2M/r_0)|} \xrightarrow{M\to 0} 4\pi r_0. \tag{12}$$

3. *As $r_0 \to 2M$, $C \to 0$ like $1/|\ln(1 - 2M/r_0)|$: a near-horizon reading certifies vanishing Killing energy.*

*Proof.* The homogeneous solutions of $(4\pi r(r-2M)\phi')' = 0$ are 1 and $u$. Finite energy excludes $u$ inside the shell; decay selects $u$ outside. Continuity and the unit jump of the flux give $G(r, r_0) = \min(u, u_0)/(8\pi M)$. A grid with harmonic conductances reproduces the matrix to $7 \times 10^{-14}$, and the innermost-node capacitance to $9 \times 10^{-14}$ at $r_{\mathrm{in}} = 2M(1 + 10^{-6})$. The flat kernel $1/(4\pi r_>)$ misses by 0.46 (canary). At $r_0 = 8M$, $C = 87.36M$ against $100.53M$ in flat space. □

**Theorem 7.2** (Transport amplitude, computed)**.** [GF] *For one shell detector at fixed areal radius $r_0$, the energy-norm amplitude of $A_{RN}$ per unit* $\mathrm{d}\ln M$ *is*

$$a(x) = \sqrt{\frac{1}{u_0}\Big[\frac{\mathcal{E}^2 - 1}{2} - \frac{(\mathcal{E}-1)^2}{u_0}\Big]}, \qquad x = \frac{2M}{r_0}, \quad \mathcal{E} = \frac{1}{1-x}, \quad u_0 = -\ln(1-x), \tag{13}$$

*with weak-field limit $a \to x/(2\sqrt{3})$ and divergence at the horizon.*

*Proof.* The representer is $\rho = \min(u, u_0)/(8\pi M)$ with $\|\rho\|_H^2 = u_0/(8\pi M)$. At fixed $r$, $\partial_M u = (e^u - 1)/M$, so $\partial_M \rho = (e^u - 1 - u)/(8\pi M^2)$ outside the shell. Projecting out $\rho$ leaves $A\rho$ with $\mathrm{d}(A\rho)/\mathrm{d}u = [e^u - (e^{u_0} - 1)/u_0]/(8\pi M^2)$ on $(0, u_0)$ and zero inside. Then $\|A\rho\|_H^2 = 8\pi M \int_0^{u_0} (\mathrm{d}A\rho/\mathrm{d}u)^2 \mathrm{d}u$ gives the formula. □

At $r_0 = 8M$, $a = 0.0962$; the grid gives 0.09643 at the nearest node $r_0 = 7.983M$, where the closed form is 0.09643. The weak-field estimate is 0.0722. Integrating $a(x)/x$ from 0 gives the transport length 0.0830 for switching the mass on at $r_0 = 8M$. The earlier hand-seeded amplitude $1 - \sqrt{1 - 2M/r_0} = 0.134$ is neither; it was an input, not a result. For four shells at $4M$, $8M$, $16M$, $30M$ the computed amplitude is 0.190.

## 8 The hole of gravity's own exterior measurement

The pipeline above concerns a field observed on a background. The same structure governs the inverse problem of gravity itself: the source is unknown and the exterior field is the data.

**Proposition 8.1** (Newtonian hole and resolution law; IDENTIFIED)**.** [GN] *Let $\rho$ be supported in a ball of radius $a$ and let the exterior potential be measured at radius $R > a$.*

1. *Every $\rho = \Delta\psi$ with $\psi$ supported in the ball has vanishing exterior field. The hole is infinite-dimensional [11].*
2. *In the $L^2$ source norm the singular value of the degree-$l$ channel is $4\pi(a/R)^{l+1}/((2l+1)\sqrt{2l+5})$ in units $a = 1$. The number of resolved multipoles is therefore about $\ln(\mathrm{SNR})/\ln(R/a)$.*
3. *The complement of the hole is not unique. With respect to the field energy $\int |\nabla\Phi|^2$ it consists of boundary layers (balayage [12]); with respect to $L^2$ it consists of harmonic densities.*

*Proof.* (1): $\int \Delta\psi\, h = \int \psi\, \Delta h = 0$ for harmonic $h$; the battery integrates $\psi_l = r^l(r(1-r))^4$ exactly for $l = 0, \ldots, 4$ and obtains 0, against 1/30 for a generic density. (2): the moment is $Q_l = \frac{4\pi}{2l+1}\int \rho_l r^{l+2}\mathrm{d}r$; the battery matches the closed form to $5 \times 10^{-5}$ and the slope $-\ln R$ to $10^{-5}$. (3): Dirichlet's principle. For unit mass the field energies are 0.0798 (shell), 0.0956 (uniform), 0.2118 (peaked), while the $L^2$ norm is least for the uniform density. □

Item (3) is complement relativity (L117) in a classical guise, and it shows what the energy form selects: the least-energy source compatible with exterior data lives on the boundary of the unresolved region.

**Proposition 8.2** (The hole in general relativity, spherical case)**.** [GO] *For static spherical bodies the exterior metric is Schwarzschild with $M = \int_0^{R_*} 4\pi r^2 \rho\, \mathrm{d}r$ (IDENTIFIED: Birkhoff; Misner–Sharp mass). The exterior map is linear in the energy density and has rank one. Its hole is $\{\delta\rho : \int r^2 \delta\rho\, \mathrm{d}r = 0\}$ within the cone of admissible interiors.*

The battery integrates the Tolman–Oppenheimer–Volkoff equation for a uniform body of radius $6M$ and a parabolic body of radius $9M$. Both have mass 1 to $10^{-10}$ and identical exterior data. Their central pressures are $1.40 \times 10^{-4}$ (matching the exact interior solution) and $6.08 \times 10^{-5}$. The exterior Jacobian has rank 1. The admissible cone has a wall: the central pressure of the uniform body diverges at the Buchdahl radius $9M/4$ [13] (battery: 0.0051, 0.045, 0.78 at $3M$, $2.4M$, $2.26M$).

## 9 The automation theorem

**Theorem 9.1** (GR automation)**.** *Let $(M, g)$ contain a static region, let $S$ be a field action whose Killing energy form $H_g$ is positive definite, and let $(B, \Sigma)$ be a detector with bounded components and fixed readings $y$. Then the following objects are determined, and each is independent of field coordinates and of admissible discretization:*

1. *the hole* $\ker B$ *and the observable quotient* $Q = V/\ker B$ *(Theorems 3.2, 4.1, 4.3);*

2. *the resolved sector* $R = \operatorname{span}\{\mathcal{G}_g b_i\}$*, the Green matrix* $G(g)$*, and the health spectrum* $\operatorname{spec}(\Sigma^{-1}G)$*, a complete invariant (Theorems 3.1, 3.5, 3.6);*

3. *the transport block* $A_{RN}[\delta g] = -P_N \mathcal{G}_g(\delta H_g) P_R$*, equal to the mixed stress matrix element, with trivial holonomy on the full sector and invariance under isometries (Theorems 5.1, 5.2, 5.4, 5.5);*

4. *the observation action* $W_{\mathrm{obs}} = \frac{1}{2} y^\top G(g)^{-1} y$*, the quotient energy, a certified lower bound (Theorems 4.1, 6.4);*

5. *the observable source* $T^{\mathrm{obs}}_{\mu\nu} = T_{\mu\nu}[\phi^\star]$*, conserved off the detector and balanced on it by the detector force (Theorems 6.1, 6.5).*

*The inputs that remain declared are* $(M, g)$*, the action* $S$ *with its improvement terms, the detector* $(B, \Sigma)$*, and, outside static regions, a positive form.*

*Proof.* Assemble the cited theorems. No step identifies the Lorentzian metric with a positive form: the metric enters through $H_g$, $\mathcal{G}_g$, the stress bilinear and the deformation tensor. □

## 10 Objections and answers

| Objection | Answer |
|---|---|
| The sectors depend on how the field is parametrized. | The spectrum is a congruence invariant and a complete one (Theorem 3.1, GA). A non-congruent change moves it (canary). |
| The numbers are artefacts of a grid. | The objects are Green-function quantities. Admissible detectors give mesh-independent limits at second order; point probes in dimension $\geq 2$ are identified as inadmissible, with the predicted logarithmic growth (Theorem 3.6, GD). |
| Capture = 1 against a random baseline proves nothing. | Correct. Both are identities (Remark 3.4). They are withdrawn as evidence. |
| The transport amplitude was put in by hand. | Correct for the earlier table. Here the block is the derivative of a projector computed from the metric, with a closed form on Schwarzschild that differs from the seeded value (Theorems 5.1, 7.2, GF). |
| The split depends on the noise model and on a threshold. | The full sector is independent of $\Sigma$. A truncated budget is a declaration, and it is the only source of holonomy at fixed detector (Theorems 3.2, 5.4). |
| A Lorentzian action has no positive Hessian. | Correct. The split fails for indefinite forms (Proposition 2.1, GM). The positive form is the Killing energy; elsewhere it is declared (Remark 2.2). |
| "Fixed resolved field" is undefined under metric variation. | Correct. The envelope identity holds for fixed readings and fails in metric-dependent frames; the general formula carries a multiplier term (Theorem 6.1, Proposition 6.2, GI). |
| $T^{\mathrm{obs}}$ is not conserved, so it cannot source the Einstein tensor. | It is conserved off the detector, and its divergence on the detector is the detector force. The detector must be included (Theorem 6.5, Corollary 6.6, GK). |

| Objection | Answer |
|---|---|
| This is known linear algebra. | Much of it is, and it is cited as such: Ky Fan, Riesz and representer theorems, the likelihood-informed subspace, Kalman, balayage, Birkhoff. What is assembled here is the chain, the identities that join its layers (Theorems 5.1, 5.2, 5.4, 6.1, 6.5), the admissibility wall, and the exact Schwarzschild example. No priority is claimed for the individual identities. |
| General relativity is not derived. | Not here. See the *Tensor Derivation* for the source tensor and the Einstein branch. |

## 11 Predictions and falsifiers

These are statements about observation on a known background. They are not new gravitational phenomena. Each can fail.

**Prediction 1** (No holonomy at full rank)**.** For a fixed detector and a full-rank sector, transport around any loop of backgrounds has trivial holonomy. *Falsifier:* a nonzero holonomy that persists when the budget is raised to rank $B$ at fixed $B$. *Use:* any reported holonomy must be attributed to truncation or to detector motion, and must vanish when both are removed.

**Prediction 2** (One-sided deficit and monotone refinement)**.** For static fields with quadratic energy, the least-energy estimate from a subset of detectors never exceeds the estimate from a superset, and never exceeds the true energy. *Test with existing laboratory physics:* electrostatic or thermal fields sampled by nested electrode or sensor sets. *Falsifier:* a decrease under refinement, which would show a non-quadratic energy or a wrong energy form.

**Prediction 3** (Capacitance law and horizon wall)**.** For a static massless scalar, a shell at areal radius $r_0$ concentric with a Schwarzschild mass has capacitance $8\pi M/|\ln(1-2M/r_0)|$. The relative correction to the flat value is $-M/r_0$ at first order, about $7\times10^{-10}$ at the Earth's surface. *Test:* numerical relativity or analogue systems with a static redshift profile. *Falsifier:* a different functional form.

**Prediction 4** (Transport amplitude)**.** Under a change of the central mass at fixed areal detector radius, the resolved sector rotates into the hole with energy-norm amplitude $a(2M/r_0)\,\mathrm{d}\ln M$. *Falsifier:* a computed or measured rotation of a different size at full rank.

**Prediction 5** (Detector force)**.** The stress inferred from static readings is piecewise conserved, with jumps equal to the forces on the detector elements. *Test:* force measurements on electrodes held at measured potentials. *Falsifier:* a nonzero divergence away from the detector.

**Prediction 6** (Multipole resolution)**.** The number of gravitational multipoles resolvable at distance $R$ from a body of size $a$ grows like $\ln(\mathrm{SNR})/\ln(R/a)$. *Falsifier:* resolution beyond this law without prior information on the interior.

## 12 Corrections to the earlier automation notes

The two notes of July 2026 (*Degenerate Forward Maps and Fisher Holes*; *Observable GR Transport and Source-Side Action Automation*) stated the pipeline and its non-claims correctly. The following points are corrected.

- *Transport blocks.* The reported blocks had norm equal to a transport amplitude that was itself an input (0.2351 and 0.1340). The agreement was an identity. Theorems 5.1 and 7.2 replace it with computed values.

- *Capture table.* Automatic capture 1.000 against random capture 0.25 at budget 25 of 100 with rank-25 data is the pair of identities of Remark 3.4.
- *Health preserved under transport.* Invariance of the health vector "in the transported frame" is congruence invariance (Theorem 3.1), not a test.
- *Envelope identity.* The frame in which the resolved field is held was unspecified. It must be the readings (Proposition 6.2).
- *Positive Hessian.* "$H_g \succ 0$" was assumed for an action Hessian. It holds for the Killing energy of a static region and fails for the Lorentzian action (Proposition 2.1).
- $\nabla^{LC} P_R$. The covariant derivative of a projector on field space was undefined. It is replaced by the derivative over the space of metrics and by the Lie derivative along flows, where the Levi-Civita connection enters through the deformation tensor (Theorem 5.5).
- *Exterior toy.* Five "moment-like" probes of a radial profile gave rank 5. In Newtonian gravity and in general relativity the exterior data of a spherical body have rank 1 (Proposition 8.2).
- *Holonomy from near-null coupling.* The observation was right. Theorem 5.4 explains it: holonomy needs truncation or a moving detector.

What stands unchanged: the elementary obstruction (null directions cannot be recovered), the separation of fit quality from identifiability, and the three-layer architecture.

## 13 Limits

- Static regions and scalar fields are treated. Stationary non-static regions, gauge fields and metric perturbations need the same analysis with constraints; they are not done here.
- Nonlinear forward maps are handled only through their linearization. The hole is then the distribution $\ker Df$, and its integrability is the constant-rank question treated in the main print.
- Outside stationary regions the positive form is declared.
- The continuum Morita statement is imported.
- The predictions are about observation structure. None is a test of general relativity against an alternative.

## A Battery

The battery `rg_gr_automation.py` needs numpy, scipy and sympy and uses seed 20261111. It writes its results to `rg_gr_automation_results.json` and exits nonzero on any failure. Twelve mutations were run against it, one or more per layer: the sign of the block formula, the flat kernel in place of the Schwarzschild one, a wrong envelope factor, a truncated sector in the full-holonomy test, the sign of the mass term in the stress, a wrong moment weight, a wrong normalization of the second interior, a positive form in the Lorentzian wall, a Gram matrix without the Green operator, a trivial group action, the sign of the mass term in the energy density, and a Fisher form that is not a coarse-graining. Each is caught by the intended group.

| Group | Content and canary |
|---|---|
| GA | congruence invariance; completeness; $\Sigma$-independence of the full sector; capture identities; non-congruence canary |
| GB | Gibbs prior: posterior, mutual information, optimal low-rank update; wrong-order and random canaries |
| GC | representers span $R$; spectrum = spec$(\Sigma^{-1}G)$; field equation with detector sources; wrong-form canary |
| GD | one-dimensional Green matrix at order $h^2$; block detector converges; point detector diverges by $\ln 2/2\pi$ per halving |
| GE | Schwarzschild Green matrix in closed form; tridiagonal inverse; capacitance; horizon wall on the grid; flat-kernel canary |
| GF | block formula and completeness; closed-form amplitude; weak-field limit; conformal massless zero block; seeded-amplitude canary |
| GG | stress bilinear against finite differences; duality; no-pressure canary |
| GH | full-sector holonomy trivial; truncated holonomy converged; retraced-path canary; Möbius detector |
| GI | envelope with fixed readings, quadratic and quartic; orthonormal-frame canary |
| GJ | energy split; quotient norm; monotone refinement; local cross stress; Euclidean-split canary |
| GK | piecewise conserved stress; jump equals detector force; global balance; null-bump canary |
| GL | isometry invariance; deformation formula in closed form; Hellmann–Feynman; Killing time; Kalman hole; non-isometry, sign and time-dependence canaries |
| GM | indefinite form: null representer, nilpotent pencil, negative relative eigenvalues |
| GN | null densities exact; resolution law; balayage against $L^2$ complement; generic-density canary |
| GO | two interiors, one exterior; rank one; Buchdahl wall |
| GP | groupoid algebra of a free action: matrix blocks over the quotient; non-free canary |
| GQ | refinement raises, coarse-graining lowers the spectrum; $W_{\mathrm{obs}}$ monotone |

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